

Mobile Communication: From 1G to 4G

Mobile communications technology has come a long way since the initial analogue phones. Read this article to understand the evolution from 1G to 4G with technologies behind this phenomenal growth and important developments along the way

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Any radio telephone capable of operating while moving at any speed, battery operated and small enough to be carried by a person comes under the mobile communication systems. These communication systems may have different facilities. The different types of mobile communication systems are mobile two-way radio, public land radio, mobile telephone and amateur (HAM) radio.

Mobile two-way radios are one-to-many communication systems that operate in half-duplex mode, i.e., push to talk. The most common among this type is citizen band (CB) radio, which uses amplitude modulation (AM). It operates in the frequency range of 26-27.1 MHz having 40 channels of 10 kHz. It is a non-commercial service that uses a press-to-talk switch. It can be amplitude-modulated having double-sideband suppressed carrier or single-sideband suppressed carrier.

Public land mobile radio is a two-way FM radio system, used in police, fire and municipal agencies. It is limited to small geographical areas.

Mobile telephones offer full-duplex transmission. These are one-to-one systems that permit two simultaneous transmissions. For privacy, each mobile unit carries a unique telephone number.

Amateur (HAM) radios cover a broad frequency band from 1.8 MHz to above 30 MHz. These include continuous wave (CW), AM, FM, radio teletypewriter, HF slow-scan still picture TV, VHF or UHF slow-scan or fast-scan

TV, facsimile, frequency-shift keying and amplitude-shift keying.

Present and past of mobile communications

Before I narrate the journey from 1G to 4G, let me explain the important technologies behind the phenomenal growth of mobile communication systems. Since the commercial introduc-



tion of advanced mobile phone system (AMPS) service in 1983, mobile communication systems have witnessed an explosive growth. The most important breakthrough was the cellular concept.

Cellular concept. The advent of cellular operation brought frequency reuse capabilities. Advances in wireless access, digital signal processing, integrated circuits, increased battery life, etc led to exponential growth of personal communication services.

Cellular system works as follows: An available frequency spectrum is divided into discrete channels, which

are assigned in groups to geographic cells covering a service area. The discrete channels are capable of being reused in different cells with diameters ranging from 2 to 50 km. The service area is allotted a radio frequency (RF) transmitter, whereas adjacent cells operate on different frequencies to avoid interference.

Cellular telephones began as a simple two-way analogue communication system using frequency modulation for voice and frequency-shift keying for transporting control and signaling information. Other cellular systems are digital cellular system, cordless telephony, satellite mobile and paging. Analogue cellular systems fall in the first-generation (1G) category and digital cellular low-power wireless fall in the second-generation (2G) category.

Analogue cellular phone. In 1970, Bell Labs in New Jersey proposed a cellular telephone concept as advanced mobile telephony system (AMPS). AMPS is a standard cellular telephone service placed into operation on October 13, 1983 by Illinois Bell. It uses narrow-band FM with a usable audio frequency band of 300-3 kHz and maximum frequency deviation of ± 12 kHz for 100 per cent modulation. According to Carson's rule, this corresponds to 30 kHz.

AMPS uses frequency-division multiple access (FDMA), where transmissions are separated in the frequency domain. Subscribers are assigned a pair of voice channels (forward and reverse) for the duration of their call. Analogue cellular channels carry both voice using FM and digital signaling information using binary FSK.

Digital cellular system. It provides improvements in both capacity and